Step 1: Load & Explore the Data

First, let's load both CSV files and inspect their structure.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the datasets
pbj_data = pd.read_csv('PBJ_Daily_Nurse_Staffing_Q2_2024.csv',
encoding='latin1')  #used encoding='latin1' to avoid
UnicodeDecodeError
nh_provider_info = pd.read_csv('NH_ProviderInfo_Feb2025.csv',
encoding='latin1')


# Display basic info
print(pbj_data.head())  # Check the first few rows
print(nh_provider_info.head())  # Check the structure of the second
dataset
```

```
C:\Users\CherRyY\AppData\Roaming\Python\Python39\site-packages\
IPython\core\interactiveshell.py:3550: DtypeWarning: Columns (0) have
mixed types.Specify dtype option on import or set low_memory=False.
  exec(code_obj, self.user_global_ns, self.user_ns)

  PROVNUM                  PROVNAME          CITY STATE COUNTY_NAME  \
0   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
1   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
2   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
3   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
4   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin

   COUNTY_FIPS  CY_Qtr  WorkDate  MDScensus  Hrs_RNDON  ...
Hrs_LPN_ctr  \
0           59  2024Q2  20240401         51      10.77  ...
0.0
1           59  2024Q2  20240402         52       8.43  ...
0.0
2           59  2024Q2  20240403         53      11.13  ...
0.0
3           59  2024Q2  20240404         52      12.27  ...
0.0
4           59  2024Q2  20240405         52       4.95  ...
0.0

   Hrs_CNA  Hrs_CNA_emp  Hrs_CNA_ctr  Hrs_NAtrn  Hrs_NAtrn_emp
Hrs_NAtrn_ctr  \
0   160.08       160.08          0.0        0.0            0.0
```

```
0.0
1   135.95       135.95          0.0        0.0            0.0
0.0
2   150.31       150.31          0.0        0.0            0.0
0.0
3   133.01       133.01          0.0        0.0            0.0
0.0
4   137.92       137.92          0.0        0.0            0.0
0.0

   Hrs_MedAide  Hrs_MedAide_emp  Hrs_MedAide_ctr
0          0.0              0.0              0.0
1          0.0              0.0              0.0
2          0.0              0.0              0.0
3          0.0              0.0              0.0
4          0.0              0.0              0.0

[5 rows x 33 columns]
  CMS Certification Number (CCN)
Provider Name  \
0                         015009                     BURNS NURSING
HOME, INC.
1                         015010               COOSA VALLEY HEALTHCARE
CENTER
2                         015012                   HIGHLANDS HEALTH
AND REHAB
3                         015014  EASTVIEW REHABILITATION & HEALTHCARE
CENTER
4                         015015                PLANTATION MANOR
NURSING HOME

              Provider Address     City/Town State  ZIP Code  \
0         701 MONROE STREET NW  RUSSELLVILLE    AL     35653
1       260 WEST WALNUT STREET     SYLACAUGA    AL     35150
2          380 WOODS COVE ROAD    SCOTTSBORO    AL     35768
3     7755 FOURTH AVENUE SOUTH    BIRMINGHAM    AL     35206
4  6450 OLD TUSCALOOSA HIGHWAY      MC CALLA    AL     35111

   Telephone Number  Provider SSA County Code County/Parish  \
0        2563324110                       290      Franklin
1        2562495604                       600     Talladega
2        2562183708                       350       Jackson
3        2058330146                       360     Jefferson
4        2054776161                       360     Jefferson

             Ownership Type  ...  \
0  For profit - Corporation  ...
1  For profit - Corporation  ...
2       Government - County  ...
3   For profit - Individual  ...
```

```
4   For profit - Individual  ...

   Number of Citations from Infection Control Inspections  Number of
Fines  \
0                                                NaN
1
1                                                0.0
0
2                                                NaN
0
3                                                0.0
0
4                                                NaN
0

   Total Amount of Fines in Dollars Number of Payment Denials  \
0                           23989.0                          0
1                               0.0                          0
2                               0.0                          0
3                               0.0                          0
4                               0.0                          0

  Total Number of Penalties
Location  \
0                          1     701 MONROE STREET
NW,RUSSELLVILLE,AL,35653
1                          0      260 WEST WALNUT
STREET,SYLACAUGA,AL,35150
2                          0        380 WOODS COVE
ROAD,SCOTTSBORO,AL,35768
3                          0   7755 FOURTH AVENUE
SOUTH,BIRMINGHAM,AL,35206
4                          0  6450 OLD TUSCALOOSA HIGHWAY,MC
CALLA,AL,35111

  Latitude Longitude  Geocoding Footnote Processing Date
0  34.5149   -87.736                 NaN      2025-02-01
1  33.1637   -86.254                 NaN      2025-02-01
2  34.6611   -86.047                 NaN      2025-02-01
3  33.5595   -86.722                 NaN      2025-02-01
4  33.3222   -87.034                 NaN      2025-02-01

[5 rows x 103 columns]
```

Step 2: Data Merging

We need to merge the two datasets based on a common identifier, likely the Provider Number.

```
# Ensure column names are aligned
pbj_data.columns = pbj_data.columns.str.strip().str.lower()
```

```
nh_provider_info.columns =
nh_provider_info.columns.str.strip().str.lower()

# Convert column names to lowercase for consistency
pbj_data.columns = pbj_data.columns.str.lower()
nh_provider_info.columns = nh_provider_info.columns.str.lower()

# Merge datasets on Provider Number (PROVNUM and CMS Certification
Number)
merged_data = pd.merge(pbj_data, nh_provider_info, left_on='provnum',
right_on='cms certification number (ccn)', how='left')

# Check first few rows
print(merged_data.head())
```

```
  provnum                    provname          city state_x county_name
\
0   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE      AL    Franklin

1   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE      AL    Franklin

2   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE      AL    Franklin

3   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE      AL    Franklin

4   15009  BURNS NURSING HOME, INC.  RUSSELLVILLE      AL    Franklin


   county_fips  cy_qtr  workdate  mdscensus  hrs_rndon  ...  \
0           59  2024Q2  20240401         51      10.77  ...
1           59  2024Q2  20240402         52       8.43  ...
2           59  2024Q2  20240403         53      11.13  ...
3           59  2024Q2  20240404         52      12.27  ...
4           59  2024Q2  20240405         52       4.95  ...

   number of citations from infection control inspections  number of
fines  \
0                                                NaN
NaN
1                                                NaN
NaN
2                                                NaN
NaN
3                                                NaN
NaN
4                                                NaN
NaN

   total amount of fines in dollars  number of payment denials  \
0                               NaN                        NaN
1                               NaN                        NaN
```

```
2                                NaN                         NaN
3                                NaN                         NaN
4                                NaN                         NaN

   total number of penalties  location  latitude  longitude  \
0                        NaN       NaN       NaN        NaN
1                        NaN       NaN       NaN        NaN
2                        NaN       NaN       NaN        NaN
3                        NaN       NaN       NaN        NaN
4                        NaN       NaN       NaN        NaN

   geocoding footnote  processing date
0                 NaN              NaN
1                 NaN              NaN
2                 NaN              NaN
3                 NaN              NaN
4                 NaN              NaN

[5 rows x 136 columns]
```

Step 3: Data Cleaning & New Metrics

```
# Fill missing contractor values with 0
merged_data[['hrs_lpn_ctr', 'hrs_cna_ctr']] =
merged_data[['hrs_lpn_ctr', 'hrs_cna_ctr']].fillna(0)

# Calculate total staffing hours (RNs, LPNs, CNAs)
merged_data['total_staffing_hours'] = merged_data['hrs_rndon'] +
merged_data['hrs_lpn'] + merged_data['hrs_cna']

# Calculate contractor percentage
merged_data['contractor_percentage'] = ((merged_data['hrs_lpn_ctr'] +
merged_data['hrs_cna_ctr']) / merged_data['total_staffing_hours']) *
100

# Define "understaffed" (staffing hours < 70% of recommended)
merged_data['understaffed'] = merged_data['total_staffing_hours'] <
(merged_data['mdscensus'] * 0.7)
```

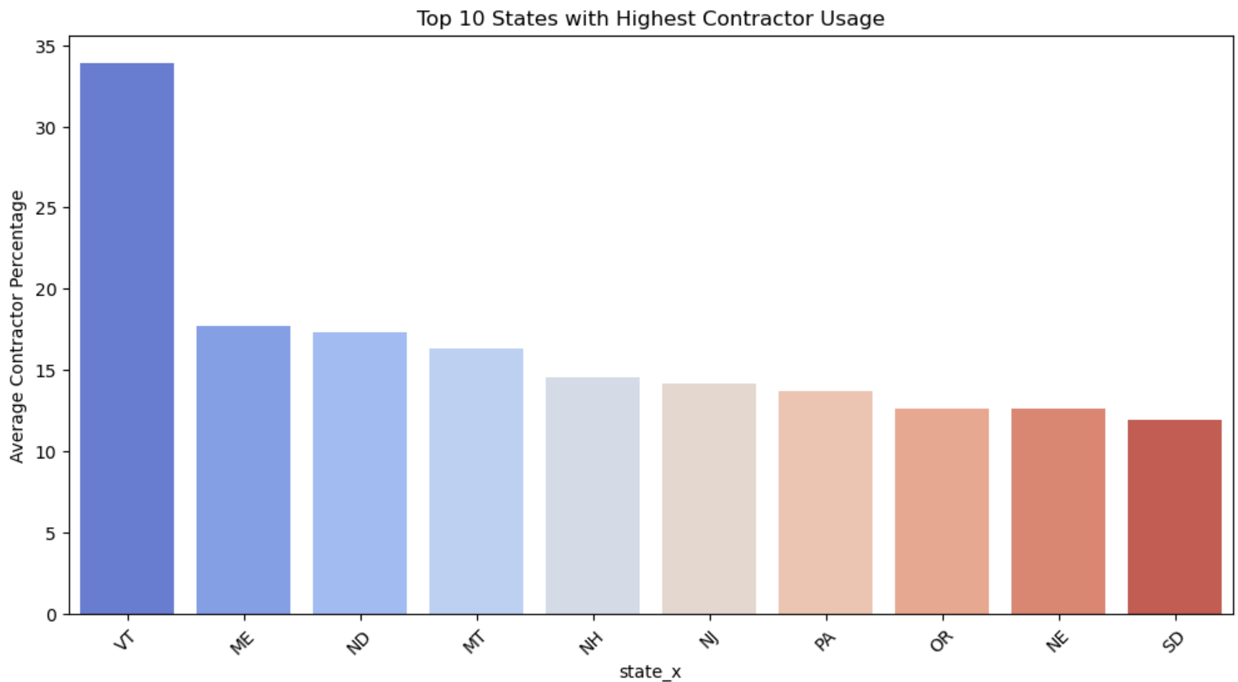
Step 4: Visualizations

1. Bar Chart: Top 10 States with Highest Contractor Nurse Usage

top 10 states with highest contractor usage percentage and sort in descending order of contractor percentage usage

```
# Group by state and calculate average contractor percentage
top_states = merged_data.groupby('state_x')
['contractor_percentage'].mean().sort_values(ascending=False).head(10)
```

```python
# Plot
plt.figure(figsize=(12, 6))
sns.barplot(x=top_states.index, y=top_states.values,
palette='coolwarm')
plt.xlabel('state_x')
plt.ylabel('Average Contractor Percentage')
plt.title('Top 10 States with Highest Contractor Usage')
plt.xticks(rotation=45)
plt.show()
```



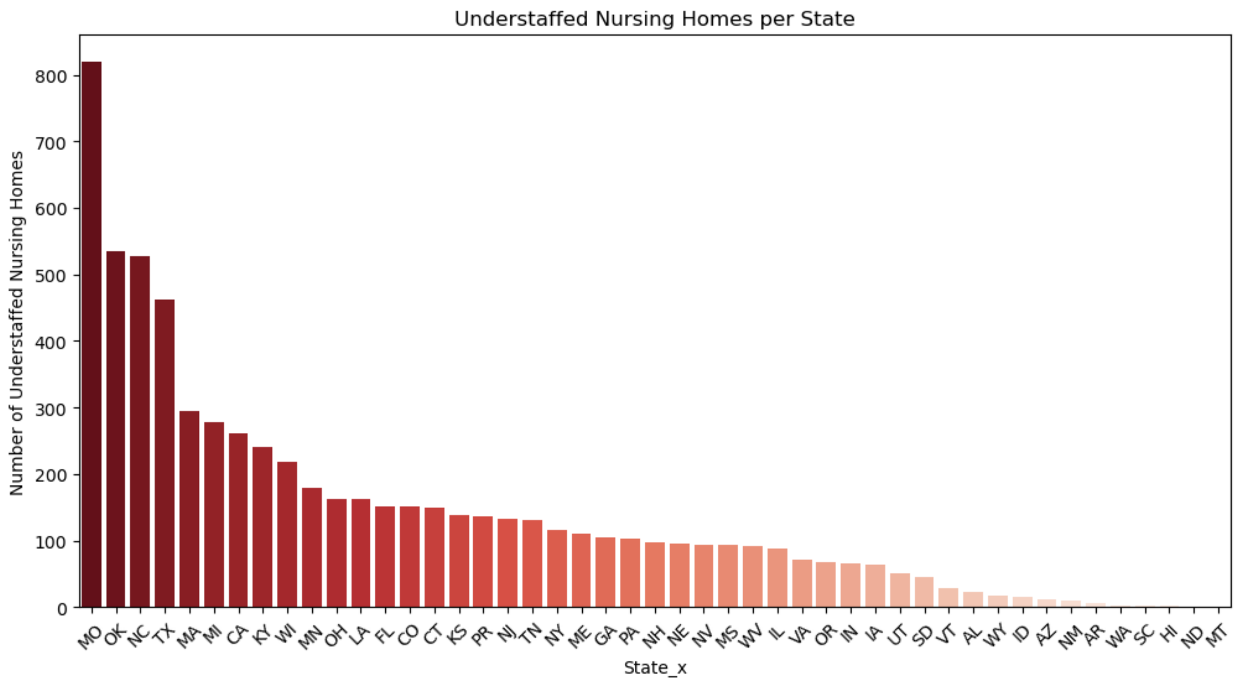
Insight: Clipboard Health should target states with high contractor usage.

2. Bar Chart: Number of Understaffed Nursing Homes per State

```python
# Count understaffed nursing homes per state
understaffed_counts =
merged_data[merged_data['understaffed']].groupby('state_x').size().sor
t_values(ascending=False)

# Plot
plt.figure(figsize=(12, 6))
sns.barplot(x=understaffed_counts.index, y=understaffed_counts.values,
palette='Reds_r')
plt.xlabel('State_x')
plt.ylabel('Number of Understaffed Nursing Homes')
plt.title('Understaffed Nursing Homes per State')
```

```
plt.xticks(rotation=45)
plt.show()
```

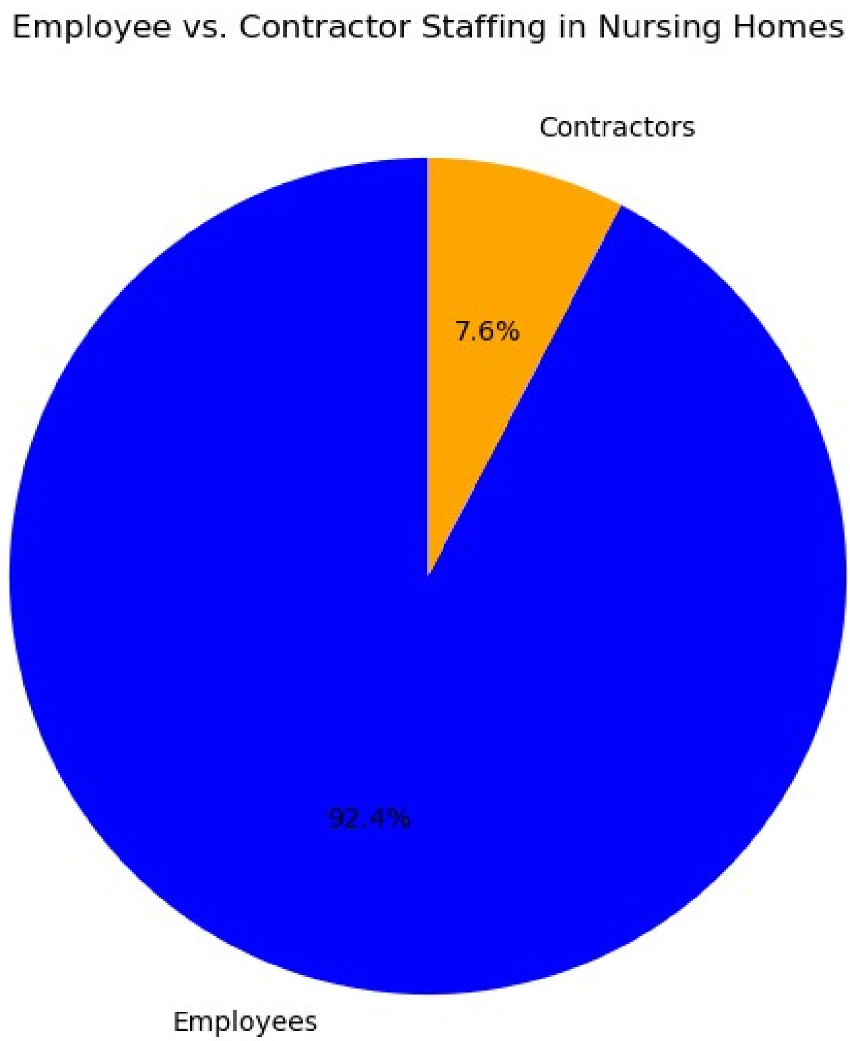


Insight: Clipboard Health should prioritize outreach to states with high understaffing rates.

3 Pie Chart: Employee vs. Contractor Staffing

```
# Calculate total employee and contractor hours
total_employee_hours = merged_data['hrs_cna_emp'].sum() +
merged_data['hrs_lpn_emp'].sum()
total_contractor_hours = merged_data['hrs_cna_ctr'].sum() +
merged_data['hrs_lpn_ctr'].sum()

# Create pie chart
plt.figure(figsize=(7, 7))
plt.pie(
    [total_employee_hours, total_contractor_hours],
    labels=['Employees', 'Contractors'],
    autopct='%1.1f%%',
    colors=['blue', 'orange'],
    startangle=90
)
plt.title("Employee vs. Contractor Staffing in Nursing Homes")
plt.show()
```



Insight: If contractor usage is low, Clipboard Health can promote contract staffing solutions.

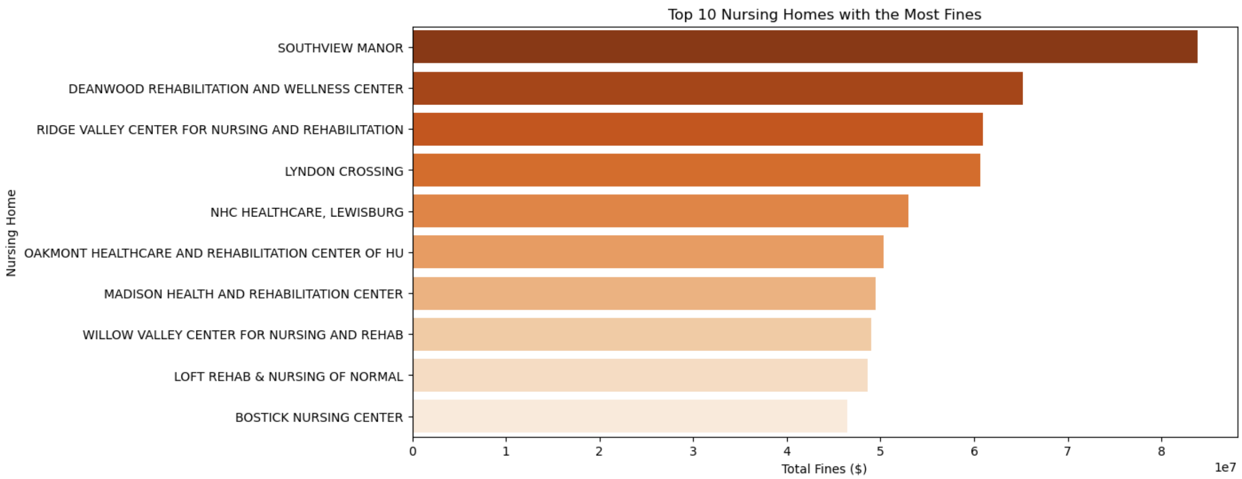
4 Bar Chart: Nursing Homes with the Most Fines

```
# Convert fine amounts to numeric
merged_data['total amount of fines in dollars'] =
pd.to_numeric(merged_data['total amount of fines in dollars'],
errors='coerce')

# Get top 10 nursing homes with highest fines
top_fines = merged_data.groupby('provname')['total amount of fines in
dollars'].sum().sort_values(ascending=False).head(10)

# Plot
plt.figure(figsize=(12, 6))
sns.barplot(x=top_fines.values, y=top_fines.index,
```

```
palette='Oranges_r')
plt.xlabel('Total Fines ($)')
plt.ylabel('Nursing Home')
plt.title('Top 10 Nursing Homes with the Most Fines')
plt.show()
```



Insight: Homes with high fines may have staffing or compliance issues—potential customers for Clipboard Health.

Key Recommendations for Clipboard Health

Here’s how Clipboard Health can leverage staffing trends to maximize impact and business growth:

1 Target High Contractor-Usage States

Some states already rely heavily on contract nurses, which means nursing homes in these areas are familiar with the model and more open to staffing partnerships. Clipboard Health can focus marketing and sales efforts here to convert existing demand into business opportunities with minimal resistance.

2 Support Understaffed Nursing Homes

Nursing homes operating with 30% or more staffing shortages are at risk of non-compliance, poor patient care, and regulatory penalties. By proactively reaching out to these facilities, Clipboard Health can position itself as a reliable staffing solution to help them meet minimum requirements and improve patient outcomes.

3 Promote Contractor Staffing to Low-Usage States

Some states still rely almost entirely on permanent staff instead of contract nurses, which may lead to burnout and long-term retention issues. Clipboard Health can introduce the benefits of flexible staffing, educating facilities on how contractors can reduce employee turnover and fill critical gaps without long hiring cycles.

4 Engage Nursing Homes with High Fines

Nursing homes that have received large fines often face issues related to staffing shortages, patient safety, and regulatory compliance. Clipboard Health can position its services as a proactive solution to help these homes avoid future penalties by ensuring they maintain adequate staffing levels.